

Week 2 Exploratory Data Analysis & Compliance Insight Report

International Student Eligibility, Enrollment Compliance, and Operational Workflow Analysis

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Week: Week 2: Exploratory Data Analysis & Compliance Insight Analysis

1. Introduction

This report presents a structured exploratory data analysis (EDA) of institutional student compliance datasets using PostgreSQL.

The analysis focused on identifying patterns related to:

- student eligibility
- enrollment compliance
- I-20 processing
- payment completion
- assignment workload distribution
- operational compliance risk indicators

SQL aggregation and segmentation techniques were used to evaluate compliance trends and identify workflow bottlenecks across institutional processes.

2. Data Preparation

The datasets were imported into PostgreSQL and analyzed using SQL-based exploratory analysis methods.

During the import process, several data preparation challenges were encountered, including:

- inconsistent CSV formatting
- delimiter conflicts

- malformed rows
- column mismatch errors
- unstructured raw text formatting

The datasets were cleaned, validated, and successfully imported into PostgreSQL before conducting compliance-focused analysis queries.

3. Eligibility Analysis

Deposit Status Distribution

The following query was used to analyze deposit completion status across students.

```
1 SELECT
2     deposit_status,
3     COUNT(*) AS total_students
4 FROM merged_data
5 GROUP BY deposit_status
6 ORDER BY total_students DESC;
```

Data Output Messages Notifications

Showing rows: 1 to 2 Page No: 1 of 1

	deposit_status text	total_students bigint
1	No	5363
2	Yes	1512

Deposit Status	Total Students
Yes	1512
No	5363

Key Observation

The majority of students have not completed deposit requirements. This may indicate enrollment-processing bottlenecks or pending financial compliance requirements.

I-20 Status Distribution

The following query was used to analyze I-20 processing completion.

```
1  SELECT
2      i20_status,
3      COUNT(*) AS total_students
4  FROM merged_data
5  GROUP BY i20_status
6  ORDER BY total_students DESC;
```

Data Output

Messages

Notifications

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	i20_status text	total_students bigint
1	No	4486
2	Yes	2389

I20 Status	Total Students
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Yes	2389
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No	4486
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Key Observation

A significant proportion of students remain without completed I-20 processing, suggesting delays in documentation or compliance workflows.

4. Enrollment Document and Eligibility Analysis

A combined analysis was conducted to identify compliance risk clusters based on deposit completion and I-20 processing status.

```
1  SELECT
2      deposit_status,
3      i20_status,
4      COUNT(*) AS student_count
5  FROM merged_data
6  GROUP BY deposit_status, i20_status
7  ORDER BY student_count DESC;
```

Data Output Messages Notifications

Showing rows: 1 to 4 Page No: 1 of 1

	deposit_status text	i20_status text	student_count bigint
1	No	No	4224
2	Yes	Yes	1250
3	No	Yes	1139
4	Yes	No	262

Deposit Status	I20 Status	Student Count
No	No	4224
Yes	Yes	1250
No	Yes	1139
Yes	No	262

Key Observation

The largest compliance segment consists of students with incomplete deposit requirements and missing I-20 processing.

Key Observation

Student assignments appear unevenly distributed across staff members.

Several records remain unassigned, while some staff combinations manage disproportionately large student caseloads.

This may indicate workload imbalance and operational dependency risks within the compliance management workflow.

6. Institutional Distribution Analysis

The dataset was analyzed based on institutional representation.

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SELECT

university,

COUNT(*) AS total_students

FROM merged_data

GROUP BY university

ORDER BY total_students DESC;

Data Output

Messages

Notifications

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SQL

	university text	total_students bigint
1	Saint Louis Univers...	6875

Key Observation

All analyzed records were associated with Saint Louis University, indicating that the current dataset represents a single institutional population.

7. Summary of Key Findings

Major Insights Discovered

- Most students have incomplete deposit requirements.
- I-20 processing completion remains relatively low.
- The largest compliance risk cluster consists of students lacking both deposit completion and I-20 processing.
- Student assignment workloads are unevenly distributed.

Notable Disparities

- Significant differences exist between completed and incomplete compliance groups.
- Large concentrations of students remain within high-risk operational categories.

Early Indicators of Compliance Risk

- High volume of unprocessed records
- Large number of incomplete compliance cases
- Uneven assignment distributions
- Operational dependency on limited staff combinations

Areas Requiring Further Monitoring

- Deposit completion progression
 - I-20 processing timelines
 - Staff workload balancing
 - Compliance backlog reduction
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8. Conclusion

The exploratory analysis successfully identified several operational and compliance-related patterns within the institutional datasets.

PostgreSQL-based analysis enabled structured segmentation of compliance status, assignment distributions, and workflow risk indicators.

The findings from this report will support the development of dashboard metrics and validation logic in subsequent project phases.